UNIVERSITY OF BOTSWANA
FACULTY OF SCIENCE
DEPARTMENT OF MATHEMATICS
MAT122 INTRODUCTORY MATHEMATICS II

2025-2026 SEMESTER 2

1. Minimum CA to Qualify for Examinations

A minimum continuous assessment (CA) mark of 40% is required to sit for final examinations.

2. LECTURERS

No.	Lecturer	Office	Email	Remark
1.	Prof. P. Kaelo	208/203	KaeloP@ub.ac.bw	**HOD**
2.	Dr. M. Koorapetse	232/238	**koorapetsem@ub.ac.bw**	**Coordinator**
3.	Dr. T. G. Motsumi	208/214	**motsumit@ub.ac.bw**	**Preparing Test 1 draft**
4.	Mr. M. Moetele	208/207	**moetelem@ub.ac.bw**	**Preparing Special Test 1 draft**
5.	Mr. P. Moile	242C/27	**moilep@ub.ac.bw**	**Preparing Test 2 draft**
6.	Dr. K. Tiro	208/208	**TiroK@ub.ac.bw**	**Preparing Special Test 2 draft**
7.	Mr. O. Ramatebele	233/127	**ramatebeleo@ub.ac.bw**	**Preparing Test 3 draft**
8.	Dr. S. Basimanebotlhe	208/210	**basimanebotlheos@ub.ac.bw**	**Preparing Special Test 3 draft**

3. LECTURE TIMES

LECTURE REGISTERED	DAY	TIME	BLOCK/ ROOM	NUMBER OF STUDENTS	LECTURER
MAT122 - LE01 (3309) MAT122_-LE01 (6356)	MoTuWeFr	0700-0800	233/G12	TBA	Dr. T. G. Motsumi
MAT122 - LE02 (3328) MAT122_-LE02 (6357)	MoTuWeFr	0700-0800	230/G5	TBA	Prof. P. Kaelo
MAT122 - LE03 (3329) MAT122_-LE03 (3411)	MoTuWeFr	0700-0800	235/003	TBA	Dr. S. Basimanebotlhe
MAT122 - LE04 (3330) MAT122_-LE04 (3412)	MoTuWeFr	0700-0800	237/036	TBA	Mr. P. Moile
MAT122 - LE05 (3331)	MoTu	1700-1800	233/G12	TBA	Mr. O. Ramatebele

MAT122_-LE05 (3413)	We	1600-1800	233/G12		
MAT122 - LE06 (3332) MAT122_-LE06 (3414)	MoTuWeFr	0700-0800	208/101	TBA	Mr. M. Moetele
MAT122 - LE07 (3333) MAT122_-LE07 (3415)	MoWeFr	0700-0800	252/002	TBA	Dr. M. Koorapetse
	Tu	0700-0800	252/001		
MAT122 - LE08 (3334) MAT122_-LE08 (3416)	MoTuWeFr	0700-0800	237/037	TBA	Dr. K. Tiro

3 Lectures + 1 Tutorial session per week

TUTORIAL GROUPS: To be formed from equitable distribution of students amongst TAs and Demonstrators

4. Course Synopsis

This course is a continuation of MAT 111 and introduces the student to some basic ideas and techniques from Calculus. It also introduces students to coordinate geometry such as conic sections. The skills obtained here will help the student appreciate the role mathematics plays in the life and physical sciences.

5. Related Exit Level Outcome(s)

LO1. Demonstrate wider knowledge and problem-solving skills in the area of Mathematics.

6. **Related Assessment Criteria**

AC1.1 Demonstrate a general understanding of the basic principles of mathematics.

AC1.2 Demonstrate knowledge through analysis of different forms of mathematical data.

7. Course Learning Outcomes

CLO1: Demonstrate understanding of basic ideas and techniques from Calculus.

CLO2: Show the ability of finding standard form of equation of a line and distance from a point

8. REFERENCES/TEXTBOOKS

- Introductory Mathematics, Customized Version for University of Botswana. Cengage Learning ISBN 978-1-4737-5405-8 (This is Main Reference Book)
- G. B. Thomas, Jr & R. L. Finney, Calculus and Analytic Geometry. Sixth Edition. Addison-Wesley Publishing Company. USA. 1984

9. SCHEME OF WORK

WEEK 1
 	
January 26-30, 2026
	CHAPTER 1. LIMITS AND CONTINUITY
	
	1.1. LIMITS
	
	• Finding Limits Graphically and Numerically
	    ✓ An introduction to limits
	    ✓ Estimating a limit using a numerical approach.
	• Evaluating Limits Analytically
	• Properties of limits
	• Evaluate a limit using properties of limits
	• The squeeze theorem
	• Evaluate limits involving $\frac{\sin x}{x}$
	• One-Sided Limits
	• Infinite Limits
	    ✓ Infinite Limits from the left and from the right
	    ✓ Vertical Asymptotes
	• Limits at Infinity

	✓ Horizontal asymptotes ✓ Infinite limits at infinity
WEEK 2 **February 02-06, 2026**	**1.2. CONTINUITY** • Definition of Continuity at a point and on an open interval • Continuity on a closed internal • Properties of continuity • Continuity of a Composite Function • The Intermediate Value Theorem
WEEK 3 **February 09-13, 2026**	**CHAPTER 2. DIFFERENTIATION** 2.1. **The Derivative of a Function** ✓ Limit definition ✓ Finding the Derivative by the limit process ✓ The derivative of the sine function ✓ The derivative of the cosine function ✓ Differentiability and Continuity 2.2. **Basic Differentiation Rules and Rates of Change** ✓ The Constant Rule ✓ The Power Rule. ✓ The Constant Multiple Rule. ✓ The Sum and Difference Rules ✓ Use derivatives to find rates of change.
WEEK 4 **February 16-20, 2026**	**2.3. Product, Quotient Rules and Higher Order Derivatives** ✓ Product Rule ✓ Quotient Rule ✓ Derivative of a trigonometric functions ✓ Higher Order Derivatives **2.4. The Chain Rule** ✓ Derivative of a composite function using the Chain Rule ✓ The General Power Rule ✓ Simplifying derivatives using algebra **2.5. Trigonometric functions and the Chain Rule**
TEST 1 SUNDAY FEBRUARY 22, 2026 @1500 HOURS	
WEEK 5 **February 23-27, 2026**	**2.6. Differentiation of logarithmic and exponential functions** ✓ Definition of the natural logarithmic function ✓ The derivative of the natural Logarithmic function ✓ The natural exponential function ✓ Derivatives of exponential functions **2.7. Implicit Differentiation**

<table>
<tr><td></td><td>✓ Implicit and Explicit functions
✓ Implicit differentiation</td></tr>
<tr><td>WEEK 6

March
02-06, 2026</td><td>2.8. Plane Curves, Parametric Equations and Calculus
✓ Plane Curves and parametric equations
✓ Eliminating the parameter
✓ Finding parametric equations
✓ Parametric Equations and Calculus
✓ Slope and Tangent lines
CHAPTER 3. APPLICATIONS OF DIFFERENTIATION
3.1 Extrema on an Interval
✓ Extrema of a function on an interval
✓ Finding extrema on a closed interval</td></tr>
<tr><td>WEEK 7

March
09-13, 2026</td><td>3.2. Increasing and Decreasing Functions and The First Derivative Test
✓ Increasing and Decreasing functions
✓ The first derivative test
3.3. Concavity and the 2nd Derivative Test
✓ Concavity
✓ Points of inflection
✓ Second Derivative Test</td></tr>
<tr><td>WEEK 8
March
16-20, 2026</td><td>MID-SEMESTER BREAK</td></tr>
<tr><td>WEEK 9

March
23-27, 2026</td><td>3.4 Applications to Curve Sketching
✓ Analyzing the graph of a function
Chapter 4. INTERGRATION
4.1. Anti-derivatives and Indefinite Integration
✓ Antiderivatives
✓ Basic Integration rules
✓ Initial Conditions and Particular solutions</td></tr>
<tr><td colspan="2">TEST 2 SUNDAY MARCH 29, 2026 @1500 HOURS</td></tr>
<tr><td>WEEK 10

March 30 –
April 03,
2026</td><td>4.2. Definite Integrals
✓ Motivation: Definite Integral as an Area under a curve
✓ Properties of Definite Integrals
✓ The fundamental Theorem of Calculus
✓ The mean Value Theorem for Integrals
✓ Average Value of a function</td></tr>
</table>

	✓ The Second Fundamental Theorem of Calculus 4.3. Integration by Substitution ✓ Pattern recognition ✓ Change of variables ✓ The General Power Rule for Integration ✓ Change of Variables for Definite Integrals ✓ Integration of Even and Odd functions ✓ Integrals of trigonometric functions
WEEK 11 **April 06-10, 2026**	4.4. Integration of Logarithmic and Exponential Functions ✓ The Natural Logarithmic Function: Integration ✓ Log Rule for Integration ✓ Inverse Functions ✓ Exponential Functions: Integration
WEEK 12 **April 13-17, 2026**	4.6. Integration by Parts ✓ Product rule in integral form. ✓ Evaluating definite integrals by parts. 4.7. Integration of rational functions by partial fractions method ✓ General description of the method
WEEK 13 **April 20-24, 2026**	**CHAPTER 5. APPLICATIONS OF INTEGRATION** 5.1 Area of a region between curves that intersect ✓ Area of a region between two intersecting curves ✓ Area of a region between intersecting curves
TEST 3 SATURDAY APRIL 25, 2026 @1500 HOURS	
WEEK 14 **April 27- May 01, 2026**	5.2. Volumes of Revolution about x-axis/y-axis
WEEK 15 **May 04- 08, 2026**	**CHAPTER 6. COORDINATE GEOMETRY** **6.1. Lines - standard forms of equations,** **6.2. Distance from a point to a line**
MAY 11-13, 2026 READING WEEK	
MAY 15-28, 2026 FINAL EXAMINATIONS PERIOD	

10. LEARNING-TEACHING STRATEGIES

10.1. Lectures

- Notes on Moodle,
- Teams or Physical presentation at scheduled times.

10.2. TUTORIALS (Demonstrators and TAs)

- Weekly Questions set by any two team members

- Tutorial groups: Equitable distribution of students amongst TAs and Demonstrators.

10.3. MATHS CLINIC (Room 233/G5) Opening hours

Working Days	Morning	**09:00-13:00**
	Afternoon	**14:00-17:00**

11. ASESSMENT

- **Three Tests ($1\frac{1}{2}$ hours each)**

  Coverage: The three tests will cover at least 90% of the contents of the course (the lectures of the last two weeks of the semester may not be covered). Lecturers decide on the coverage of each of the tests and moderate them to ensure that each test has equal share of the contents of the course.
- **Written Examination (**2 Hours)

  Coverage: All course contents except under special circumstance
- Category Weight: **CA= 50** **EXAM=50**
- **Final Mark**

Raw marks:	Average of 3 tests (CA)…..……………...	100%
	Examination (EXAM)……………………	100%

Final Mark = (CA + EXAM)/2